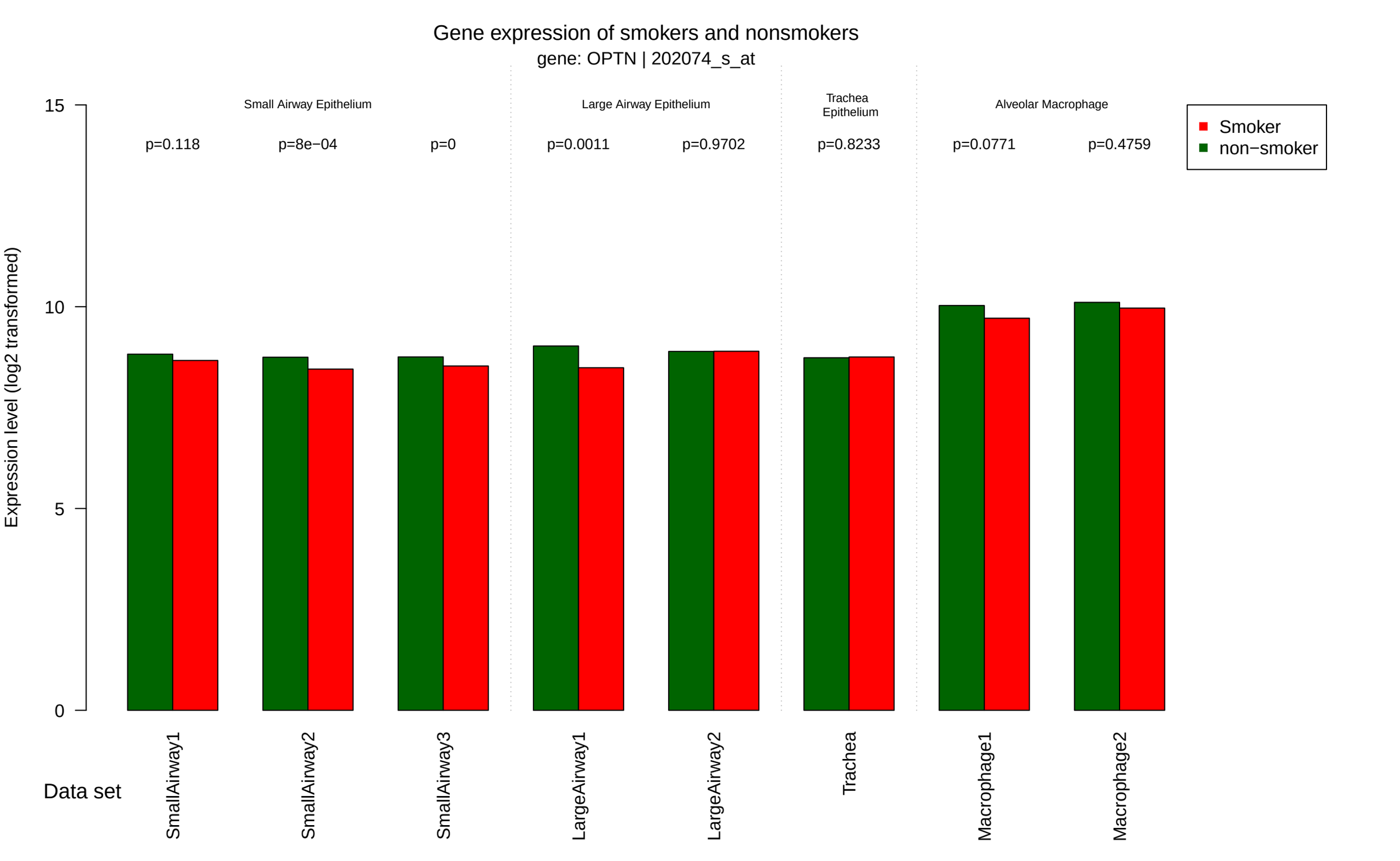
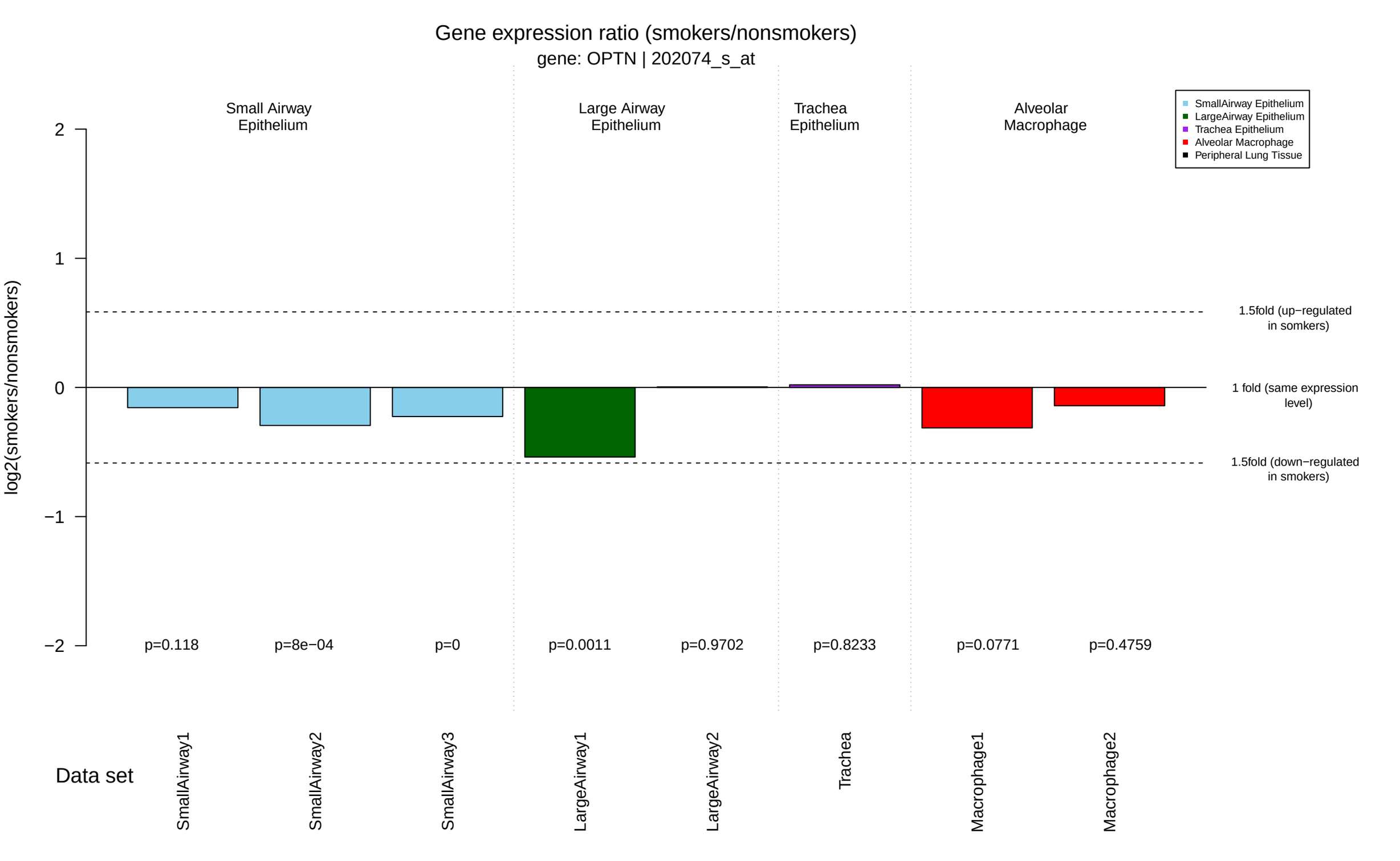
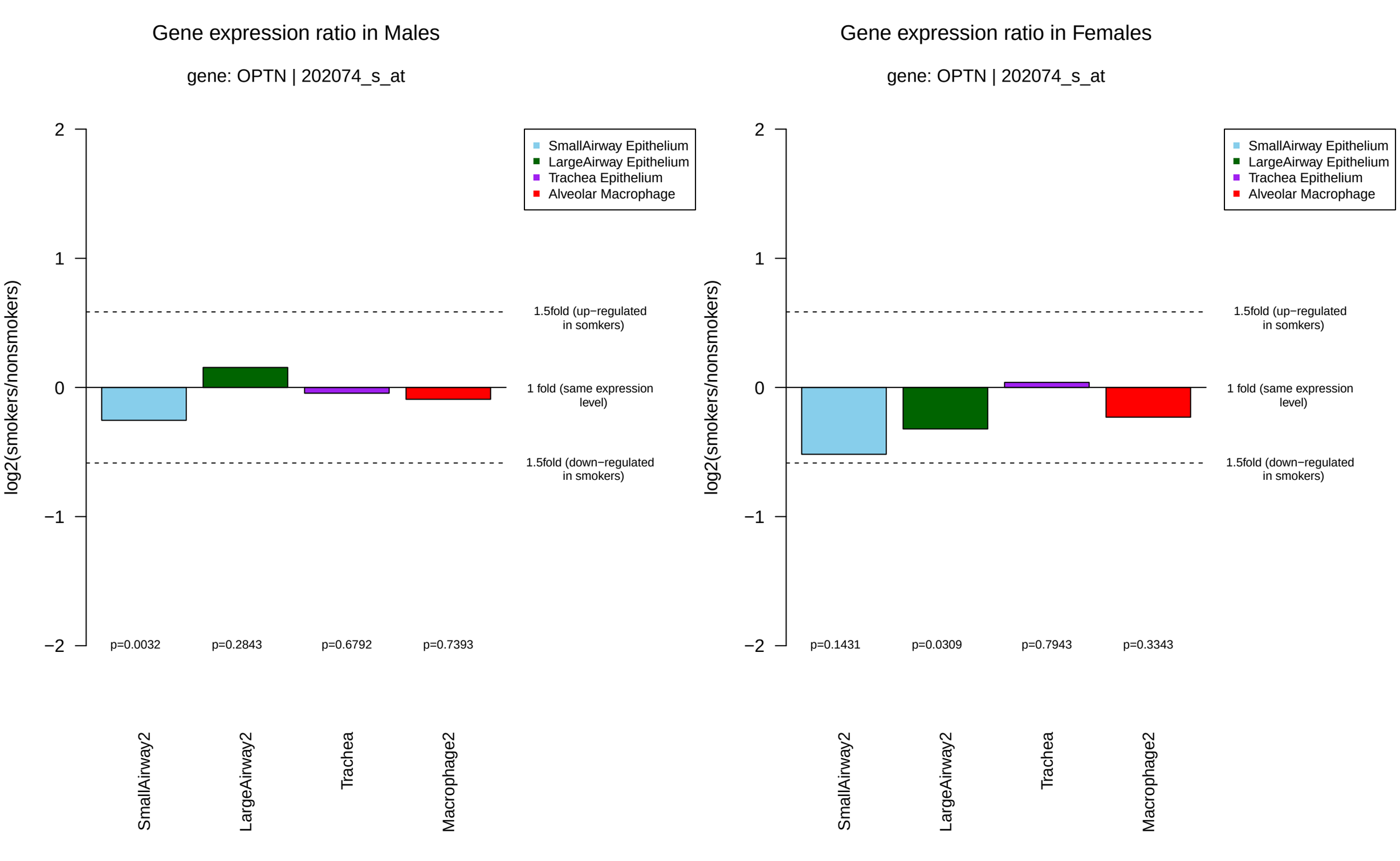
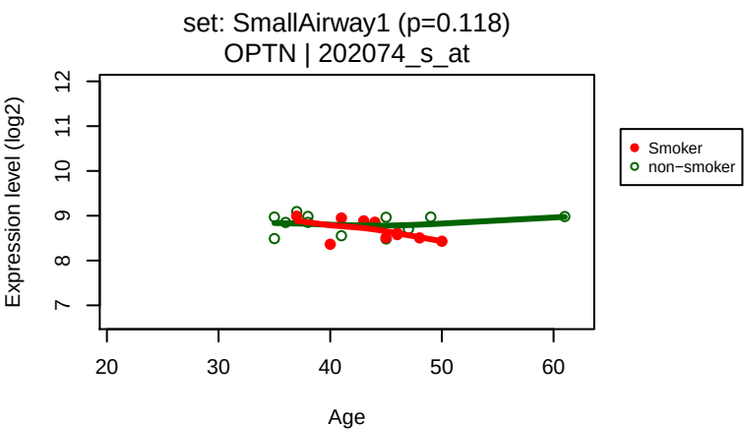
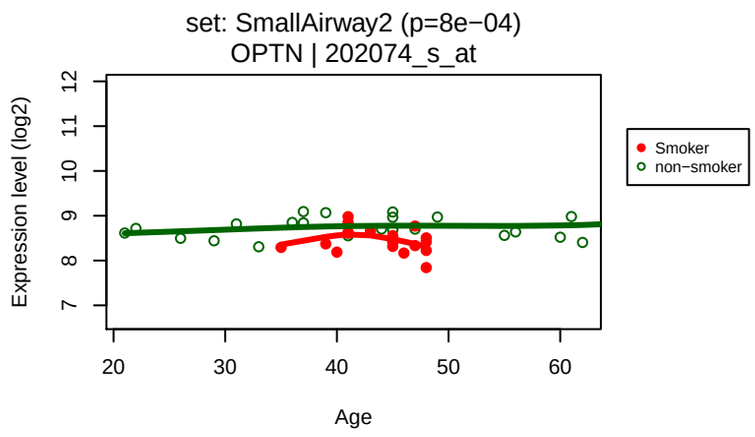
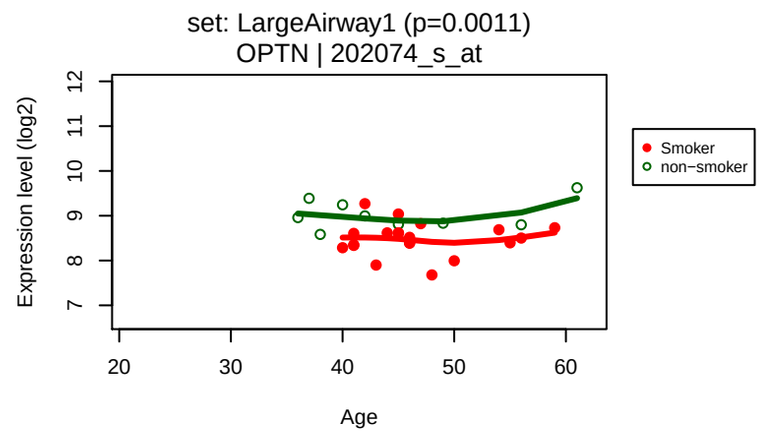

Gene expression of smokers and nonsmokers
gene: OPTN | 202074_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.118
p=8e−04
p=0
p=0.0011
p=0.9702
p=0.8233
p=0.0771
p=0.4759
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

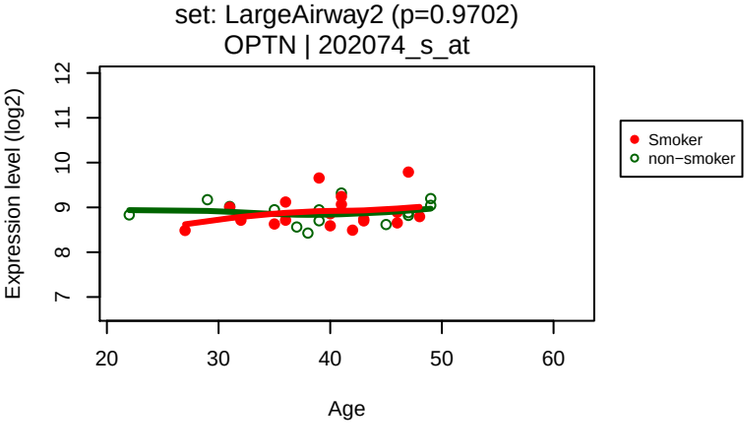

Gene expression ratio (smokers/nonsmokers)
gene: OPTN | 202074_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.118
p=8e−04
p=0
p=0.0011
p=0.9702
p=0.8233
p=0.0771
p=0.4759
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

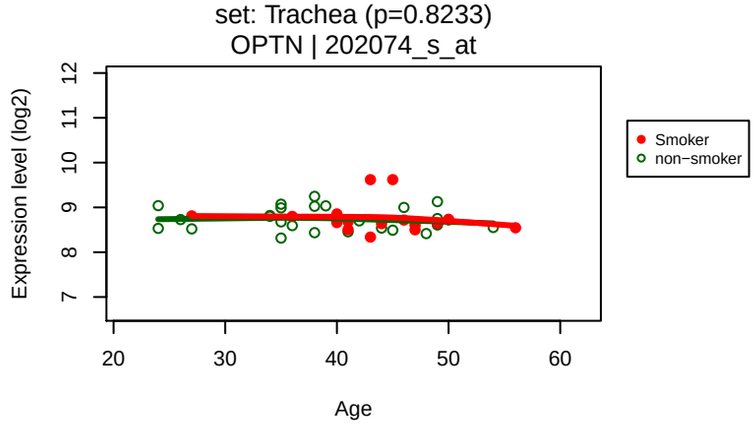

Gene expression ratio in Males
gene: OPTN | 202074_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0032
p=0.2843
p=0.6792
p=0.7393
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OPTN | 202074_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1431
p=0.0309
p=0.7943
p=0.3343
SmallAirway2
LargeAirway2
Trachea
Macrophage2

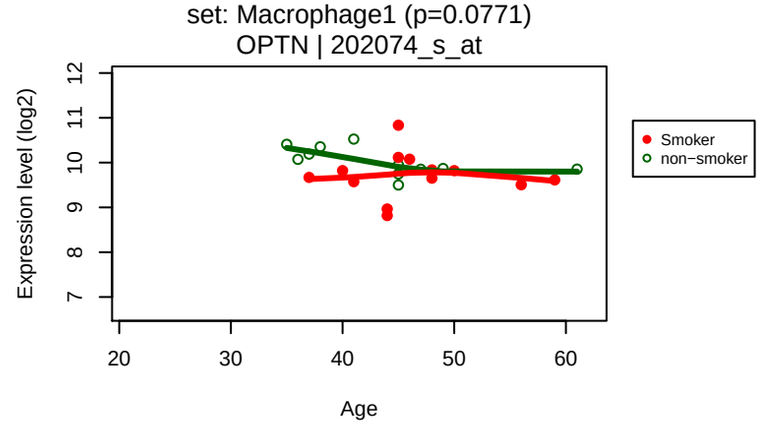

set: SmallAirway1 (p=0.118)
OPTN | 202074_s_at
Expression level (log2)
Age
Smoker
non−smoker

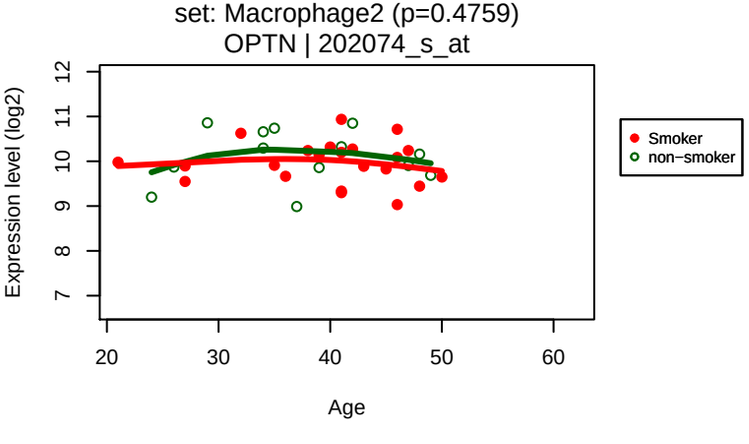

set: SmallAirway2 (p=8e−04)
OPTN | 202074_s_at
Expression level (log2)
Age
Smoker
non−smoker

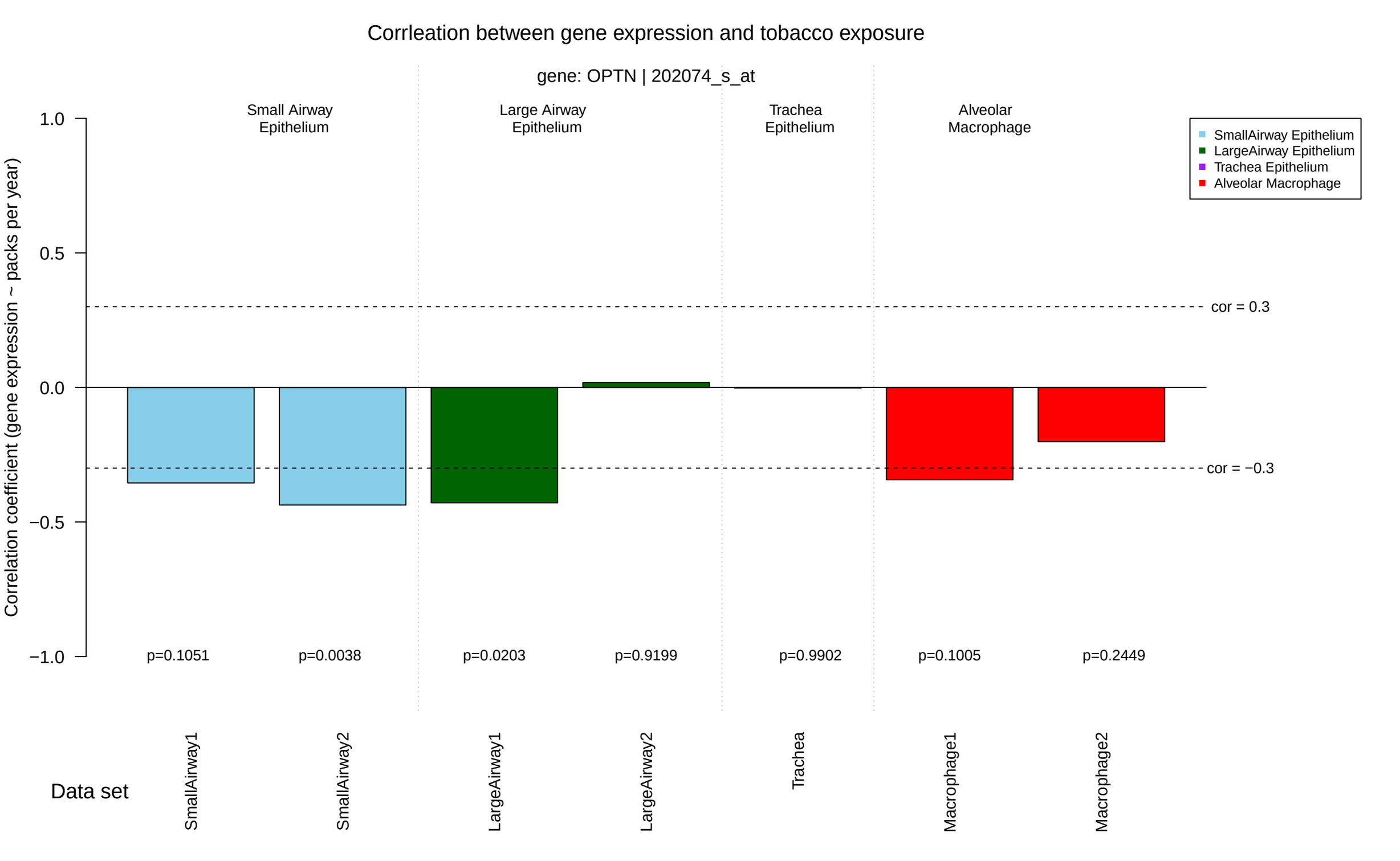

set: LargeAirway1 (p=0.0011)
OPTN | 202074_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9702)
OPTN | 202074_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8233)
OPTN | 202074_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0771)
OPTN | 202074_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4759)
OPTN | 202074_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OPTN | 202074_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1051
p=0.0038
p=0.0203
p=0.9199
p=0.9902
p=0.1005
p=0.2449
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2